



POLITECNICO
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Tutoring Section 2 Class 2

Numerical Problems in Machining

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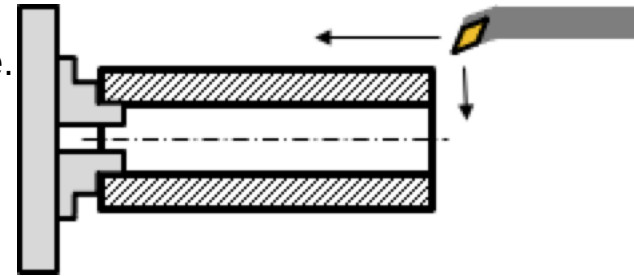
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MACHINING : TURNING OPERATIONS

■ Problem 3 (05/Feb/2021 exam)

The finishing of a hollow cylindrical (tubular) steel component is carried out with the same tool by performing two operations (see figure):

- I. External cylindrical turning over the entire length of the workpiece.
- II. Facing the free base (right).



Using the following data, determine:

1. The cutting speed to maximize the production rate.
2. The maximum allowed feed to meet a surface finish requirement on the outer cylindrical surface equal to $R_a = 0,4 \mu\text{m}$.
3. The minimum clamping pressure to properly hold the workpiece (the feed rate is $0,08 \text{ mm/rev}$, a self-centering three-jaws chuck is used with jaw-workpiece contact area equal to 120 mm^2 and friction coefficient equal to $0,35$).
4. The contact time of the facing operation in the hypothesis of carrying out the machining with a constant number of revolutions per minute equal to 540 and a feed rate of $0,08 \text{ mm/rev}$. And determine the value of the maximum cutting speed in this situation.

Data: Initial external diameter = 236 mm , final external diameter = 233 mm , initial length = 300 mm , internal diameter = 150 mm , $k_{c0,4} = 2100 \text{ MPa}$, $x = 0,29$, tool changing time = 120 s , tool radius = $0,8 \text{ mm}$, $k_r = 100^\circ$, $k_r' = 10^\circ$, Taylor's law coefficient: $C = 510$, $n = 0,35$.



MACHINING : TURNING OPERATIONS

■ Problem 4 (21/Jun/2021 exam)

In the figure, the final geometry of a mechanical component is represented. To reach that shape, an external finishing operation is made on a lathe, mounting the workpiece between centers, and considering the following data: $a_p = 1$ mm, $k_{cs} = 2100$ MPa, $x = 0,29$; $\kappa_r = 65^\circ$, $\kappa'_r = 65^\circ$; component length equal to 90 mm.

1. Considering $n = 3200$ RPM, determine the maximum and the minimum cutting speeds.
2. Considering $n = 3200$ RPM and $f = 0,15$ mm/rev., determine the cutting force and power to machine the external cylindrical surface.
3. The company estimates for each machined component a tool contact time of 40 s and a loading / unloading time of 30 s. It is planned to use tool inserts with two cutting edges each, which guarantee (under current cutting conditions) a tool life per cutting edge equal to 4,5 minutes. The tool (cutting edge) changing time is equal to 30 s. The cutting edge cannot be replaced during a component machining, and, at the end of the shift, the last insert mounted on the tool body is discarded even if it has a residual life sufficient for other machining operations or a completely new cutting edge. Calculate the production time per part (T_p), the number of parts that can be machined in an 8-hour shift, and the total cost of tool inserts in an 8-hour shift knowing that the unit cost is 8 €/insert.

